

HARP-Select: Validation-Calibrated Selection of Complementary Residual Graph Experts

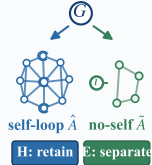
Fixed Graph Split

fixed masks for both experts



Structural Views

same nodes and features



Parallel residual graph experts

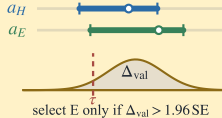
identical labels, masks, optimizer, and training budget

STRUCTURE FILTER BANK NODE FIELD FUSION VAL. SCORE



Validation evidence

validation labels only



Mechanism zoom: complementary residual bases

H: residual node fields

$$L_{k-1} - L_k = H_k$$

$$L_k = \hat{A}^k X, \quad H_k = L_{k-1} - L_k$$

E: no-self propagation and fusion

$$B_0 \rightarrow B_k = \tilde{A}^k B_0 \rightarrow [B_0, U_i]$$

explicit ego anchor + filtered neighbors

Frozen choice and test

route first, evaluate second

